PeriGen, Inc.

PeriWatch Vigilance® System Requirements

Version 2019 Rev. 1.02

PeriGen
Advanced
Perinatal Systems

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Enterprise Model (Multi-Facilities) Server Requirements

- Single server VMs providing the following main functions:
  - Application Services & Decision Support
  - SQL Server DB
  - HL7 ADT and results Interface
  - Report services

- See Certificate Requirement section for configuration of communication security
- System sizing is based on peak utilization and as a function of total monitored areas. Sizing below is based on projected requirements for approximately 5 years.
- Final configuration and optimization based on actual enterprise layout will be determined at system implementation planning stage
- VMware is currently supported on version 6.x. and above

PeriWatch Vigilance – Enterprise Application Server (VM)

For multi-server configuration and SQL clustered/Farmed configuration, please contact the PeriGen technical team

Up to # of Monitored locations	50	100	200
# of vCPUs	4	6	8
GB RAM	12	16	20
Volume C: OS	100 GB	100 GB	100 GB
Volume D: Applications	40 GB	40 GB	40 GB
Volume E: SQLData	260 GB	330 GB	400 GB
Volume F: SQLTransactionLog	50 GB	50 GB	50 GB
Volume G : SQLBackup	260 GB	330 GB	400 GB

- Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
- Microsoft Windows Server 2012 R2 or Windows 2016
  - Internet Information Services (IIS) 8.0
  - ASP.NET
  - Windows Authentication
- MS SQL 2014 SP2 Standard or SQL 2016 Standard SP2 with Reporting Services
- Windows server required modules on this and each server mentioned below:
  - Microsoft Web Services Enhancements (WSE) 3.0
  - VC++ 2010 SP1 Redistributable, VC++ 2013 Redistributable, VC++ 2015 Redistributable
  - .Net 3.5 SP1, .Net 4.7.1
  - .Net Core Windows Hosting 2.2.0

NOTE: Final sizing to be determined during pre-install implementation planning.

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NOTE: Mixed-mode enterprise data acquisition configurations (example: One site using Network Redirectors while the other is using DIGI terminal servers), will require an additional small VM per acquisition mode. The main enterprise server can accommodate one mode only. For more information, please contact the PeriGen technical team.

PeriWatch Vigilance – Mixed-Mode Data Acquisition VM (if required)

- 2 vCPUs
- 6GB RAM
- Storage requirements per volume:
  - C: Operating system - 100 GB
- Gigabit Ethernet
- Microsoft Windows Server 2012 R2
- Windows server required modules as mentioned on the primary server

PeriWatch Vigilance – Test server (required)

- 4 vCPUs
- 8GB RAM
- Storage requirements per volume:
  - C: Operating system - 100 GB
  - D: Applications/SQLData/Logs - 200 GB
- Gigabit Ethernet
- Microsoft Windows Server 2012 R2 or Windows 2016
  - Internet Information Services (IIS) 8.0
  - ASP.NET
  - Windows Authentication
- MS SQL 2014 SP2 Standard or 2016 Standard SP2 with Reporting Services
  Windows server required modules as mentioned on the primary server

Certificate Requirements

- The customer must provide a server-side SSL certificate to encrypt all data in transit across the system.
- The customer is responsible for all certificate renewal management.
- A certificate for SQL TDE is optional for full DB encryption on SQL Enterprise or Data Center versions.
- The SSL certificate requirements are:

Signature Algorithm:	SHA256RSA	
Signature hash algorithm:	SHA256	
Subject:	<FQDN of server>	(ex: VigilanceApp01.Hospital.Org)
Subject Alternative Name:	<FQDN of server>	(ex: VigilanceApp01.Hospital.Org)
Public Key:	RSA (2048 Bits)	

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Key Usage:	Digital Signature, Key Encipherment (a0)
Enhanced Key Usage:	Server Authentication (1.3.6.1.5.5.7.3.1)
	Client Authentication (1.3.6.1.5.5.7.3.2)

For SQL TDE, the certificate must have the AT_KEYEXCHANGE set to 1

Client / Workstation Requirements

- Main Vigilance client is fully browser based. Pop-up blockers should be disabled for the Vigilance application main URL or server needs to be added to the "Trusted Servers" group in GPOs. Optional thick client modules (PeriWatch Vigilance Tracings Client for administrative tasks or Vigilance Notification Agent) requirements are listed below.
- JavaScript must be enabled on client devices
- Microsoft Windows 7/10 Professional 32/64-bit. 64 bit recommended.
- 4GB RAM. 500 MB of available memory should be reserved for PeriWatch Vigilance.
- 1024x768 (4:3) is the minimum supported resolution but it is not recommended
- 1280x1024 or higher wide screens resolutions are recommended
- Native resolutions only. non-native resolutions or aspect ratios will result in a stretched representation of fetal tracings and are not supported
- If the administrative clients are installed on the station:
    - .NET Framework 4.7.1
    - VC++ 2010 SP1 Redistributable, VC++ 2013 Redistributable
- Internet Explorer 11 web browser minimum
- Google Chrome recommended, MS Edge supported

Thin-Client Technologies

- PeriWatch Vigilance HUB is fully browser based and supported on any thin-client technology (Citrix, VMware View).
- Validation steps of this product are to be executed on the customer specific version(s) of thin-client infrastructure deployed at the customer site. This is part of standard system installation activities.
- Citrix server / VMware view server sizing is a function of expected load profile. Recommendations will be finalized during system implementation.

Network (For Workstations and Data Acquisition)

- Switched Ethernet network, 100 Mbit or better

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Tracing Data Acquisition

PeriGen provide a proprietary, secure and resilient solution dedicated to collect data from existing terminal server hardware used by the existing surveillance system:

For sites using DIGI ConnectPort or PortServer hardware, PeriGen provide a Network Redirector (NR) hardware to mirror tracing network traffic. Network Redirector has the following requirements:

- POE capable switch to power the NR device
- The NR must be connected to the same switch where the third-party DIGI device is connected
- NR must have two ports available on the said switch. One for receiving one for sending.
- Only the NR's receiving port requires POE. The sending port does not require POE and can be connected to another switch in case the primary switch is full.
- Customer needs to configure "port mirroring" from the port where the third-party system DIGI device is connected to the port where the NR is connected.

For sites using MOXA terminal server, com port mirroring must be enabled on the MOXA device directly to allow a parallel data acquisition. For other configurations, please contact the PeriGen technical or support team at 1-855-236-4353, support@perigen.com

Fetal Monitor Compatibility

Vendor	Protocol compatibility	Type	Interface card / Firmware	Description
Philips	1371	* 1350A and B * 50XM series	M1350-66532 or M1350-66536 * Firmware: **C01.03 and above**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
Philips	1371	Avalon FM20 Avalon FM30 Avalon FM40 Avalon FM50	Firmware: **F.0X and above.**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
GE/Corometrics	1371	118	J09-J10	DB9 RS232 serial interface
GE/Corometrics	1371	120/170/250 Series	J09-J10-J11	RJ11 RS232 serial interface

* Philips 1350 A/B and 50XM are presently out of support by Philips. Although the serial protocol is still supported, PeriGen cannot be held responsible for known issues with these fetal monitors with firmware versions lower than **C01.03.** Known issues include a very high frequency of SP02 measurements sent to the electronic tracing when the paper recorder door is open, and no paper is in the-0 tray.

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Network ports used by the system and accessories devices

	Direction relative to device	Source	Destination	Protocol	App Protocol	Port	PHI	Description
Core system services	Bi-directional	DAS, Workstations	App Server	TCP	HTTPS	443	Yes/ Secure	Mandatory. All Communications with the Repository back-end
	Bi-directional	DAS	App Server	TCP	Proprietary	8000	No	Only if DAS is used. Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Workstations	App Server	TCP	Proprietary	8002	No	Mandatory. Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Workstations	App Server	TCP	Proprietary	8004 (See 8014)	Yes	Mandatory. Send / Receive Messages from Clients to be broadcast to Workstations
	Bi-directional	Workstations	App Server	TCP	Proprietary	8014	Yes / Encrypted	Mandatory. Send / Receive Messages from Clients to be broadcast to Workstations
Ports used when Patterns/Cues are installed	Bi-directional	Decision Support Server	App Server	TCP	Proprietary	7101	Yes / Encrypted	Only when Cues/Patterns is purchased/installed. Data communication to the decision support modules
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	7802	Yes / Encrypted	Only when Cues/Patterns is purchased/installed. Used for tracings interpretation data feed
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	7803	Yes /Encrypted	Only when Cues/Patterns is purchased/installed. Plugins service, Cues algorithm data feed
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	8081	Yes / Secured	Only when Cues/Patterns is purchased/installed. Data feed from decision support modules
	Bi-directional	Workstations	Decision Support Server	TCP	SignalR	8082	No / Secured	Only when Cues/Patterns is purchased/installed. Live data feed from decision support modules
Ports used when the Vigilance Platform is installed	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	8083	Yes / Secured	Only when Vigilance is purchased/installed. Vigilance dashboard – single patient browser-based app
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	8084	Yes / Secured	Only when Vigilance is purchased/installed. Vigilance HUB multi-patient browser-based app
	Bi-directional	Decision Support Server	Decision Support Server	TCP	Net.tcp	8201	Yes / Secured	Only when Vigilance is purchased/installed. HUB DataCollector, patient information & Cues algorithm data feed
	Bi-directional	Decision Support Server	Decision Support Server	TCP	Net.tcp	8203	Yes / Secured	Only when Vigilance is purchased/installed. HUB DBService, patient information & Cues algorithm data feed
Ports used when DIGI/MOXA terminal servers are used	Bi-directional	Network Redirectors	App Server	TCP	Proprietary	8012	No	ONLY IF DIGI SERIAL SERVERS ARE USED. Sends data acquired from the Physiological monitors over the TCP connection
	Bi-directional	Moxa Nport 6000 Series	App Server	TCP	Secure Shell	22	No	Device configuration, secured command line interface
	Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	TCP	HTTP	80	No	Device configuration web UI
	Bi-directional	Moxa Nport 6000 Series	App Server	TCP	HTTPS	443	No	Secure Device configuration web UI

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Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	TCP	Proprietary	950-965	No	Sends data acquired from the Physiological monitors over the TCP connection
Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	TCP	Proprietary	966-981	No	Command Port
Bi-directional	Moxa Nport 5000 Series, 6000 Series	App Server	UDP	Proprietary	4800	No	Broadcast search, Monitoring the status of the NPort, getting the current settings, and Real COM mode mapping
Bi-directional	Moxa Nport 5000 Series	App Server	TCP	Proprietary	4900	No	Device Firmware Upgrade Port
Bi-directional	Moxa Nport 6610-32	App Server	TCP	Proprietary	982-997	No	Sends data acquired from the Physiological monitors over the TCP connection
Bi-directional	Moxa Nport 6610-32	App Server	TCP	Proprietary	998-1013	No	Command Port
Bi-directional	Digi	App Server	TCP	Proprietary	771	No	RealPort Service; Sends data acquired from the Physiological monitors over the TCP connection
Bi-directional	Digi	App Server	TCP	Proprietary	1027	No	Secure RealPort Service; Sends data acquired from the Physiological monitors over the TCP connection
Bi-directional	Digi	App Server	TCP	Secure Shell	22	No	Device configuration, secured command line interface
Bi-directional	Digi	App Server	TCP	HTTP	80	No	Device configuration web UI

Note: The above list outlines all possible ports with the PeriWatch system (across all possible modules). Always confirm with the PeriGen technical consultant on the applicable ports for your precise configuration.

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